



JKBOSE

Baaz
Batch
Zindabad.

🔥 2026 CLASS 11th 🔥

**SOME BASIC CONCEPT
OF CHEMISTRY**

CHEMISTRY

LECTURE-1

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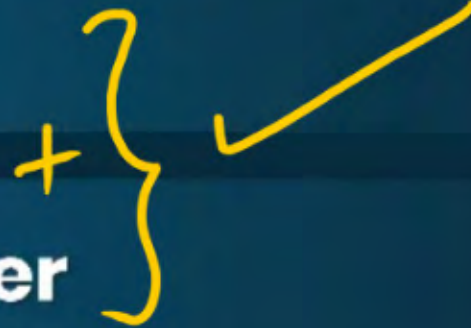


TOPICS TO BE COVERED

1 Importance of chemistry

2 Nature of matter

3 Classification of matter



JKBOSE SYLLABUS



UNIT-1: SOME BASIC CONCEPTS OF CHEMISTRY ✓

07 MARKS

General Introduction: Importance of studying chemistry, Historical approach to particulate nature of matter, Laws of Chemical combination (numerical), Dalton's Atomic Theory, Concept of elements, atoms & molecules. Atomic and molecular masses, Mole concept and molar mass, percentage composition, empirical and molecular formula; chemical reactions, stoichiometry and calculation based on stoichiometry.

IMPORTANCE OF CHEMISTRY

Definition: Chemistry is that branch of science, that deals with the preparation, properties, structure and reactions of any substance.

IMPORTANCE OF CHEMISTRY

Few important applications of chemistry are.

1) Health Sector :→ Chemistry has provided us many life saving drugs.

example :→ Taxol & Cis-platin (Anti-cancer drug), AZT → Azidothymidine (Used in the treatment of AIDS)

Others like Analgesics, Antibiotic, Anaesthetics etc

2) Warfare :→ Chemistry has provided many deadly weapons or compounds e.g

Dynamite, Nitroglycerine, TNT etc.

Others are atom bomb, hydrogen atom etc.

3) Food & Supply :→ As we know population is growing day by day. Therefore food production should be very high.

Chemistry has provided us fertilisers like urea, NH_4NO_3 etc.

Crop is also saved from various diseases, insects by insecticide, pesticides, weedicides etc.

Chemistry has also provided us preservative, that helps in preserving

the food for longer time.

4) Atomic & Nuclear energy :→

Sources of energy, Coal, petroleum, Natural are depleting day by day. Chemistry has provided alternate source of energy atomic & Nuclear energy.

5) Saving the Environment :-

Chemistry has replaced the CFC gas used previously in refrigerators, & was harmful for ozone layer. But now it is replaced by environment friendly gas HFC.

6) Transport :-

Almost all means of transport run on petrol, diesel, CNG etc all are chemicals.

7) Development of Nation :-

Chemistry has made super conducting ceramic, conducting polymers (provides no resistance) & optical fibres (no signal loss).

TRICK

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Health Sector

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Warfare

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Food

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Atomic & Nuclear energy

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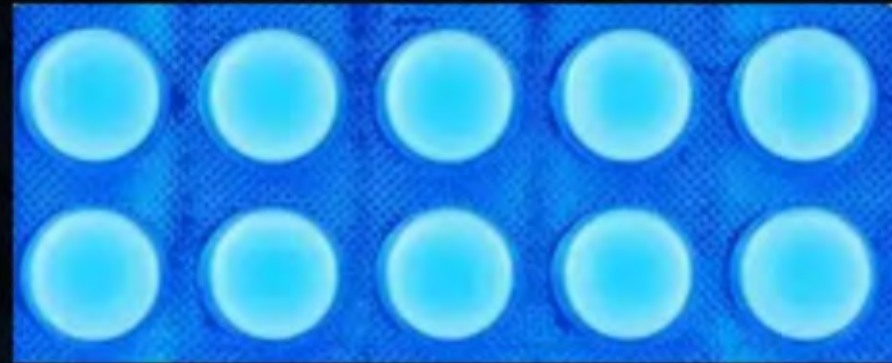
Saving the environment

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Transport

N

Nation



QUESTION

Which of the following methods can separate the components of a mixture?

- 1** Hand-picking
- 2** Filtration
- 3** Distillation
- 4** All of the above



QUESTION

At what conditions can the states of matter interconvert?

- 1** Change in color
- 2** Change in temperature and pressure
- 3** Change in volume
- 4** Change in texture



Thank
You

Jab bhi
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